GPN2-40.5kV Cubicle Type Gas Insulated Switchgear

Summary

Cubicle type gas insulated switchgear(CGIS) is an indoor, factory-assembled, metal-enclosed, cubicle type gas-insulated switchgear for single busbar applications, including "Green Type" GPN2E-40.5, GPN2N-40.5 and "Standard Type" GPN2S-40.5.

The "Green type" GPN2N-40.5 is innovated to use pure nitrogen as insulation gas for the series product with Non-SF_6 gas insulation technology, which has brought the genuine green environmental protection of gas insulated switchgear.

The "Green type" GPN2E-40.5 incorporates the advanced technologies of mixed gas-insulated (SF_6+N_2) and vacuum breakers, allowing the equipment to operate in a more reliable and environmentally friendly manner.

The "Standard type" GPN2S-40.5 is of 100% SF_6 insulated, high performance and easy use.

With modern digital manufacturing and automatic testing coupled with sensor, monitor and protection technology, CGIS is ideal fit for power distribution demands. CGIS is particularly suited to industries with high reliability requirements such as Power Networks, Mining, Rail Transportation, Petrochemical Plants, Wind Farms and Metropolitan Rail Systems.

Normal operating conditions

The switchgear is fundamentally designed for the normal service conditions for indoor switchgear to GB 3906, DL/T404 and IEC 62271-200. The following limit values, among others, apply:

Ambient air temperature

Maximum air temperature: +45℃

Minimum air temperature: -25℃

Daily average maximum temperature: +35℃

Humidity:
Daily average value of relative humidity: ≤ 95%
Monthly average value of relative humidity: ≤ 90%
Daily average value of water vapor pressure: ≤ 2.2×10^{-3}MPa
Monthly average value of water vapor pressure: ≤ 1.8×10^{-3}MPa

Altitude: ≤ 1000m

The ambient air is not significantly polluted by dust, smoke, corrosive and/or flammable gases, vapours or salt.

Special service conditions

The product can also be applied for many special service conditions.

In case the service conditions exceed the normal service conditions, which are out of the standard GB 11022 and IEC 62271-1, please consult with GP in advance for confirmation:
Altitude higher than 1000m.
Higher environmental temperature.
Lower environmental temperature.
Others special environmental conditions.

Reduce greenhouse gas emissions

CGIS incorporates the advanced technologies of pure Nitrogen or mixed gas-insulated (SF_6+N_2) and vacuum breakers, a fundamental choice made by GP to assist in reducing the greenhouse effect. SF_6 (sulphur-hexafluoride) is on the list of greenhouse gasses in the Kyoto Protocol, with a Global Warming Potential (GWP) of 23,000. Many other medium voltage switchgear systems use SF_6 gas as the only insulating medium. Leakage of SF_6 gas from switchgear contributes to the threat of the greenhouse effect and associated climate change.

With our commitment to protection of the environment, CGIS helps reduce greenhouse gas emissions by utilising mixed gas-insulated technology together with vacuum switching technology.

A 100% or 50% reduction in SF_6 is achieved by using Nitrogen (N_2) mixed gas-insulated breakers. Nitrogen is the largest component of air and its arc decomposition product is non-toxic. Joined together by plug-in connectors and the modular nature of the panels ensures ease of installation and extension without the need for extra gas handling activities on-site.

Advantage

- **Maximum safety for operator and equipment**

  The minimum functional pressure of the cubicle is 0.00MPa(20℃). That means, even under such severe conditions, it still maintains the rated insulation level and keeps all its rated functionality. Thanks to the low pressure of gas, even if gas is escaping from the switchgear, the cubicle can still continue to be energized. Reliable electrical and mechanical interlocks are designed between the circuit breaker and three-position switch to prevent misoperation.

- **70% Reduction in switchroom size**

  An optimized electric field design combined with excellent insulating performance, results in a compact switchgear product that operates safely and reliably.

  Save 70% space compared with air-insulated switchgear.

  Retrofitting into existing switchroom is easy.

  Reduce cost of substation land.

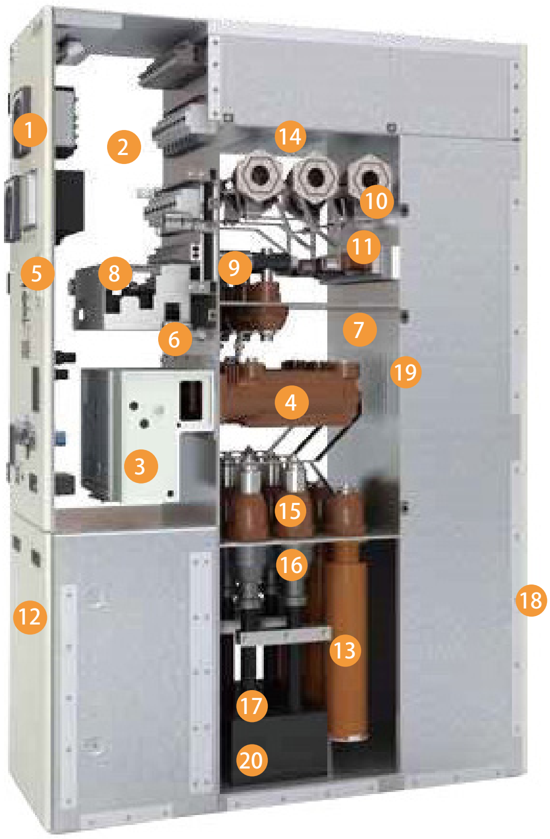
- **Easy installation / Low operation and maintenance cost**

  Panels in the middle can easily be removed for maintenance without moving neighbouring panels, increasing availability.

Technical parameters

General		Unit	Standard type GPN2S-40.5	Green Type GPN2E-40.5	Green Type GPN2N-40.5
Rated voltage		kV	36/38/40.5	36/38/40.5	36/38/40.5
Rated power frequency withstand voltage (1min)	To earth/phase to phase	kV	95	95	95
	Across isolating distance	kV	118	118	118
	To earth/phase to phase	kV	185	185	185
	Across isolating distance	kV	215	215	215
Rated frequency		Hz	50/60	50/60	50/60
Rated current		A	1250, 2500, 3150	1250, 2500	1250, 2500
Single capacitor bank breaking capacity		A	400/400	400/400	400/400
Rated cable charging breaking current		A	50	50	50
Rated shortcircuit breaking current		kA	20/25/31.5	20/25/31.5	31.5
Rated short circuit making current(peak)		kA	50/63/80	50/63/80	80
Rated short time withstand current and endurance time		kA/s	20/3, 25/3, 31.5/3s	20/3, 25/3, 31.5/3s	31.5/3s
Rated peak withstand current		kA	50/63/ 80	50/63/ 80	80
Operating sequence			O-0.3s-CO-180s-CO	O-0.3s-CO-180s-CO	O-0.3s-CO-180s-CO
Gas system insulated gas			100%SF_6	50%SF_6+50%N_2	100%N_2
Annual leakage rate		%/Y	≤ 0.1	≤ 0.1	≤ 0.1
Rated gas pressure (abs, 20˚C)		MPa	0.12	0.12	0.12
Alarm pressure (abs, 20˚C)		MPa	0.11	0.11	0.11
Minimum operating pressure (abs, 20˚C)		MPa	0.10	0.10	0.10
Degree of Protection					
Gas tank			IP65	IP65	IP65
Enclosure			IP4X	IP4X	IP4X
Motor and trip coil					
Circuit breaker charging motor		W	90	90	90
Rated power of closing coil		W	220	220	220
Rated power of opening coil		W	220	220	220
Rated voltage of auxiliary circuit		V	DC 24, 48, 110, 220; AC220	DC 24, 48, 110, 220; AC220	DC 24, 48, 110, 220; AC220
1 min power frequency withstand voltage of auxiliary circuit		kV	2	2	2
Dimensions and Weight					
Dimension (W × D × H) 1250A		mm	600 × 1600 × 2400	600 × 1500 × 2400	800 × 1700 × 2300
Dimension (W × D × H) 2500A		mm	800 × 1600 × 2400	800 × 1500 × 2400	900 × 1700 × 2300
Weight 1250A		kg	800 ~ 1000	800 ~ 1000	800 ~ 1000
Weight 2500A		kg	1100 ~ 1400	1100 ~ 1400	1100 ~ 1400

Structure of the standard type GPN2S-40.5 and green type GPN2E-40.5

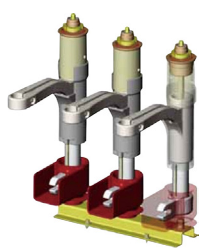


Standard type GPN2S-40.5kV

Green type GPN2E-40.5kV

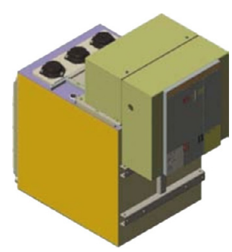
1. Protection and control Unit
2. Low voltage compartment
3. VCB mechanism
4. Embedded pole vacuum circuit breaker
5. Low voltage compartment door
6. Gas density indicator
7. VCB gas tank
8. 3-position switch mechanism
9. 3-position switch
10. Main busbar
11. Main busbar gas tank
12. Front cover
13. Surge arrester
14. Pressure relief device of main busbar gas tank
15. Inner-cone cable bushing
16. Cable terminal
17. Cables
18. Rear cover
19. Pressure relief device of VCB gas tank
20. CT

Standard type:
GPN2S-40.5 VCB



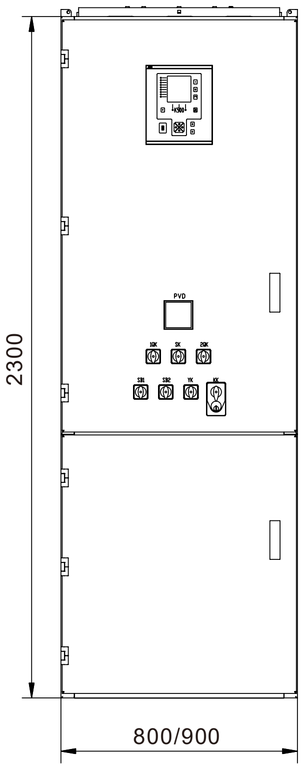
IST 3-position mechanism

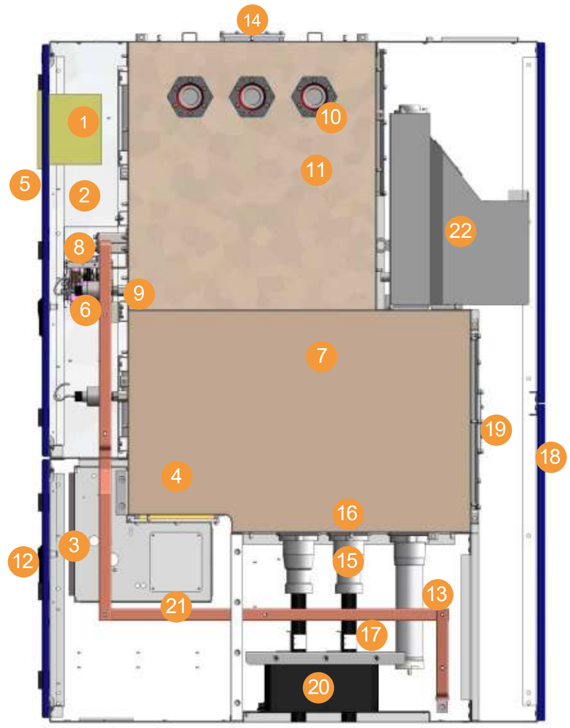
IST 3-position switch



GPN2S VCB gas tank

Structure of the GPN2N green type (No-SF_6)





Green type GPN2N-40.5kV

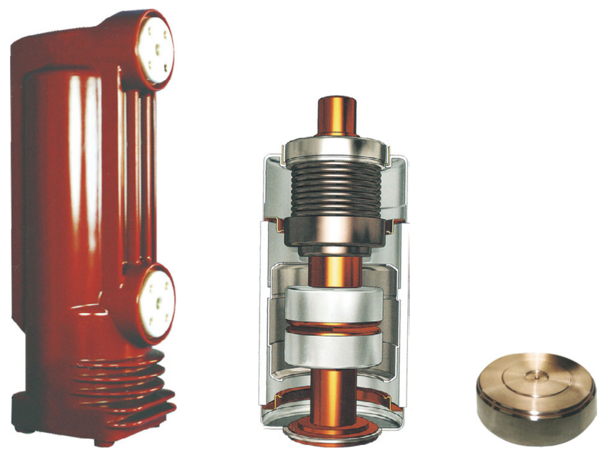
1. Protection and control unit
2. Low voltage compartment
3. VCB mechanism
4. Embedded pole vacuum circuit breaker
5. Low voltage compartment door
6. Gas density indicator
7. VCB gas tank
8. 3-position switch mechanism
9. 3-position switch
10. Main busbar
11. Main busbar gas tank
12. Front cover
13. Surge arrester
14. Pressure relief device of main busbar gas tank
15. Inner-cone cable bushing
16. Cable terminal
17. Cables
18. Rear cover
19. Pressure relief device of VCB gas tank
20. CT
21. Earthing bar
22. Voltage transformer (optional)

Vacuum circuit breaker is fixed mounted, while its three-phase embedded poles are arranged vertically into the circuit breaker gas tank.

Due to the vacuum switching technology, an arc is limited in the vacuum interrupter, reducing the exhaust volume of insulation gas. Vacuum switching is of high performance in frequent short-circuit and numerous on-load switching applications.

PT installation

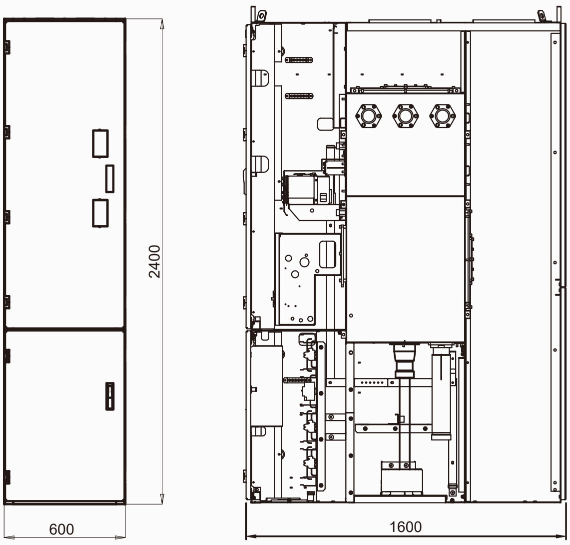
CT in cable compartment



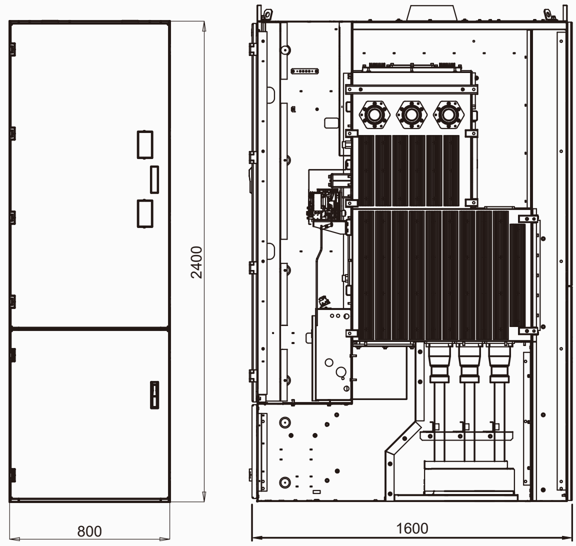
The vacuum interrupter of the VCB is of an optimized design, the ceramic insulator is compact with high insulation level and a high current breaking capability. The contacts are made of Copper-Chromium, with excellent abrasion resistance, long electrical endurance, and high short circuit breaking capacity.

The embedded poles are assembled on the flange plate of the frame. The vacuum interrupter and terminals are fixed inside the embedded pole by APG epoxy molding process.

Outline dimension of GPN2S-40.5kV standard type

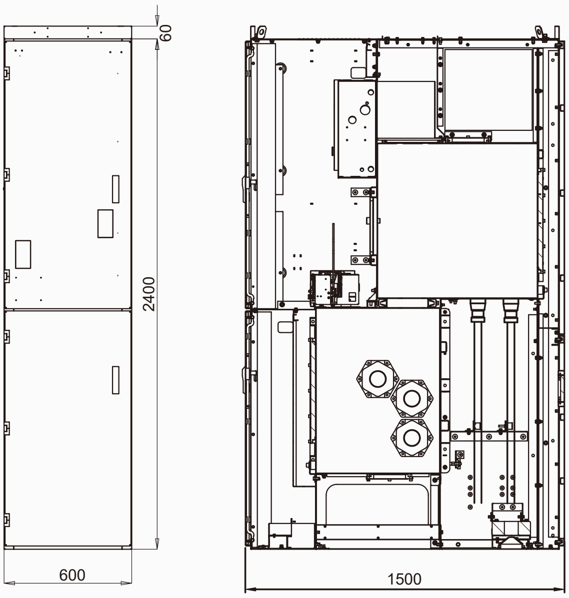


GPN2S-40.5kV-1250A Standard type

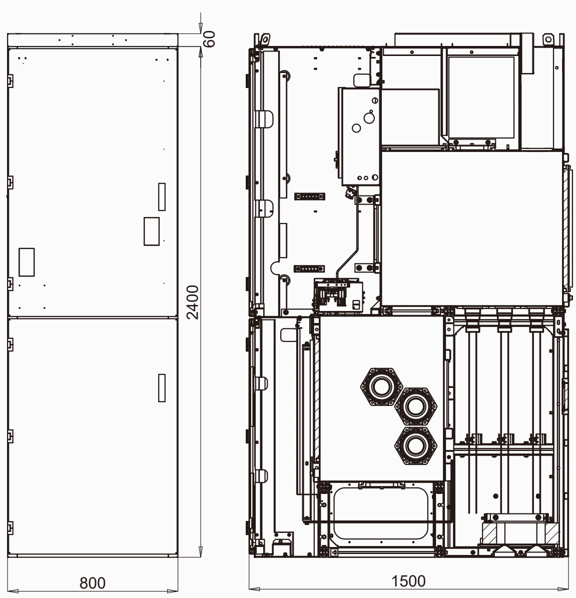


GPN2S-40.5kV-2500A~3150A Standard type

Outline dimension of GPN2E-40.5kV green type



GPN2E-40.5kV-1250A Green type



GPN2E-40.5kV-2500A Green type